Program 1: Introduction to Maven and Gradle: Overview of Build Automation Tools, Key Differences Between Maven and Gradle, Installation and Setup

Overview of Build Automation Tools

Build automation tools help developers streamline the process of building, testing, and deploying software projects. They take care of repetitive tasks like compiling code, managing dependencies, and packaging applications, which makes development more efficient and error-free.

Two popular tools in the Java ecosystem are **Maven** and **Gradle**. Both are great for managing project builds and dependencies, but they have some key differences.

Maven

- Maven is a build automation tool primarily used for Java projects. It uses an XML configuration file called `pom.xml` (Project Object Model) to define project settings, dependencies, and build steps.
- **Main Features:**
  - Predefined project structure and lifecycle phases.
  - Automatic dependency management through Maven Central.
  - Wide range of plugins for things like testing and deployment.
  - Supports complex projects with multiple modules.

Gradle

- Gradle is a more modern and versatile build tool that supports multiple programming languages, including Java, Groovy, and Kotlin. It uses a domain-specific language (DSL) for build scripts, written in Groovy or Kotlin.
- **Main Features:**
  - Faster builds thanks to task caching and incremental builds.
  - Flexible and customizable build scripts.
  - Works with Maven repositories for dependency management.
  - Excellent support for multi-module and cross-language projects.Integrates easily with CI/CD pipelines.

Installation and Setup

Java (JDK) Installation:

Step 1: Type the command **sudo apt install openjdk-11-jdk -y in the terminal**

It will install the OpenJDK 17 (Java Development Kit) and accepts installation automatically.

```
sumanth@sumanth-VirtualBox: ~
sumanth@sumanth-VirtualBox:~$ sudo  apt install openjdk-11-jdk -y
[sudo] password for sumanth:
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
openjdk-11-jdk is already the newest version (11.0.30+7-1ubuntu1~22.04).
0 upgraded, 0 newly installed, 0 to remove and 88 not upgraded.
sumanth@sumanth-VirtualBox:~$
```

The installation will start like this.

Step 2: Installation can be confirmed by running the command **java -version**

```
sumanth@sumanth-VirtualBox:~$ java -version
openjdk version "11.0.30" 2026-01-20
OpenJDK Runtime Environment (build 11.0.30+7-post-Ubuntu-1ubuntu122.04)
OpenJDK 64-Bit Server VM (build 11.0.30+7-post-Ubuntu-1ubuntu122.04, mixed mode, sharing)
sumanth@sumanth-VirtualBox:~$
```

Maven Installation:

Step 1: Type the command **sudo apt install maven** in the terminal.

It will install Apache Maven from Ubuntu's package repository.

```
sumanth@sumanth-VirtualBox:~$ sudo apt install maven
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
maven is already the newest version (3.6.3-5).
0 upgraded, 0 newly installed, 0 to remove and 88 not upgraded.
```

Step 2: Installation can be confirmed by running the command **mvn -version**

```
sumanth@sumanth-VirtualBox:~$ mvn -version
Apache Maven 3.6.3
Maven home: /usr/share/maven
Java version: 11.0.30, vendor: Ubuntu, runtime: /usr/lib/jvm/java-11-openjdk-amd64
Default locale: en_IN, platform encoding: UTF-8
OS name: "linux", version: "6.8.0-106-generic", arch: "amd64", family: "unix"
```

<u>Gradle Installation:</u>

Step 1: Type the command **sudo apt install gradle** in the terminal

It installs Gradle build automation tool from Ubuntu's package repository.

```
sumanth@sumanth-VirtualBox:~$ sudo apt install gradle
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
gradle is already the newest version (4.4.1-13).
0 upgraded, 0 newly installed, 0 to remove and 88 not upgraded.
sumanth@sumanth-VirtualBox:~$
```

Step 2: To check the proper installation of Gradle run the command **gradle –version**

```
sumanth@sumanth-VirtualBox:~$ gradle -version
WARNING: An illegal reflective access operation has occurred
WARNING: Illegal reflective access by org.codehaus.groovy.reflection.CachedClass (file:/usr/share/java/groovy-all.jar) to method jav
a.lang.Object.finalize()
WARNING: Please consider reporting this to the maintainers of org.codehaus.groovy.reflection.CachedClass
WARNING: Use --illegal-access=warn to enable warnings of further illegal reflective access operations
WARNING: All illegal access operations will be denied in a future release

------------------------------------------------------------
Gradle 4.4.1
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Build time:   2012-12-21 00:00:00 UTC
Revision:     none

Groovy:       2.4.21
Ant:          Apache Ant(TM) version 1.10.12 compiled on January 17 1970
JVM:          11.0.30 (Ubuntu 11.0.30+7-post-Ubuntu-1ubuntu122.04)
OS:           Linux 6.8.0-106-generic amd64

sumanth@sumanth-VirtualBox:~$
```